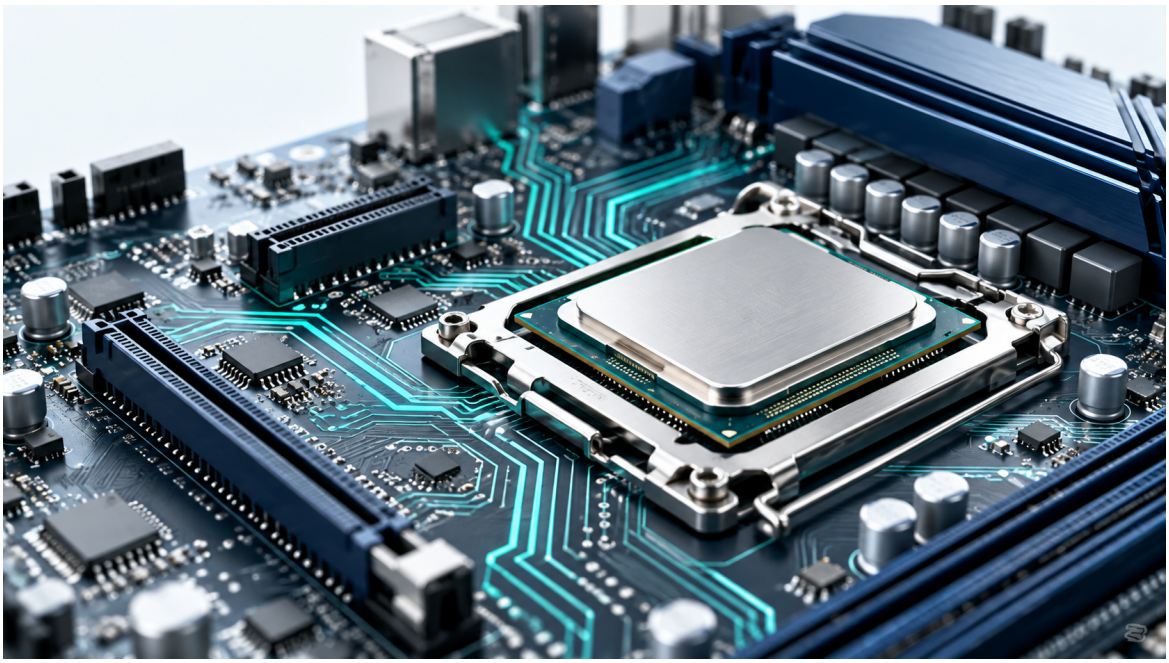
Operating Systems

Basics se CPU Scheduling tak

Visual Hinglish Study Guide



Core idea: OS applications aur hardware ke beech baithkar resources manage karta hai. CPU scheduling decide karti hai ki ready process ko CPU kab milega.

Computer system ki layers

USER
Aap aur aapke apps

APPLICATIONS
Browser, editor, game, database

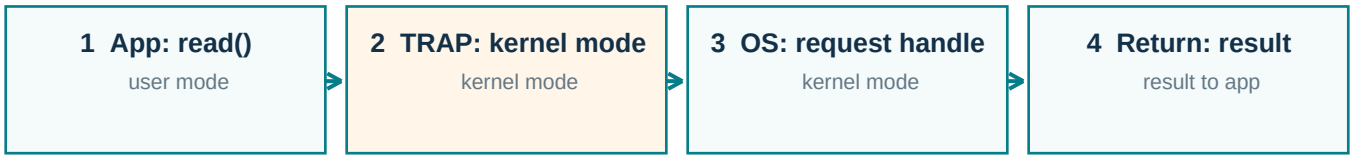
OPERATING SYSTEM
Interface + resource manager

HARDWARE
CPU, memory, disk, keyboard, display

Jaise restaurant mein waiter customer aur kitchen ke beech request le jaata hai, waise OS app ki request hardware tak pahunchata hai. OS ka goal: computer ko convenient banana aur resources ko efficiently use karna.

System call: app kernel se service maangti hai

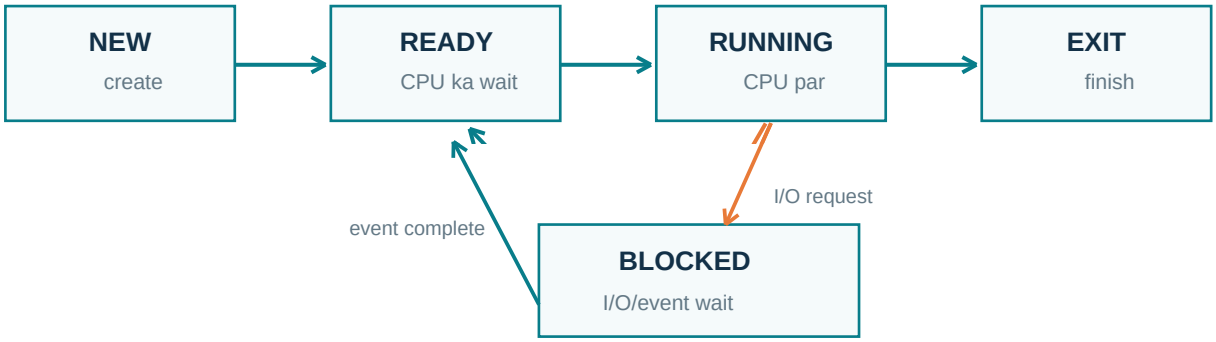
App ko file read karni ho toh woh hardware ko seedha command nahi deti. System call ke through OS se request karti hai. TRAP CPU ko controlled tareeke se kernel mode mein le jaata hai.



Mode switch = privilege badalta hai. Context switch = CPU par chalne wala process badalta hai. Dono ek hi cheez nahi hain.

Process ki life cycle

Process disk par program hota hai; execution shuru hote hi OS uska PCB banata hai. PCB mein state aur saved CPU details rehti hain.

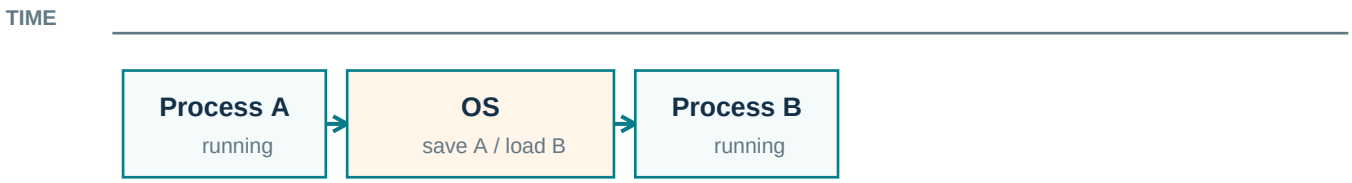


Ready = run karne ko taiyar, CPU ka wait. Blocked = I/O ya event ka wait.
Event complete hone par Blocked process wapas Ready queue mein aata hai.

Schedulers: long-term jobs admit karta hai; short-term ready queue se CPU chunta hai; medium-term processes ko memory aur disk ke beech swap karta hai.

Context switch: ek process se doosre par

Timer interrupt, I/O request ya higher-priority process aane par OS switch kar sakta hai. Pehle current process ke registers/PC ko PCB mein save karta hai; phir next process ka saved state restore karta hai.



Context switch is overhead: switch ke dauraan app ka useful work nahi hota. Isliye bahut zyada switching performance ghata sakti hai.

Scheduling ke do choices

NON-PREEMPTIVE

CPU process ke paas rehta hai jab tak finish ya I/O wait na ho. Example: FCFS, SJF.

PREEMPTIVE

OS running process se CPU le sakta hai. Naya/shorter job aaye toh switch ho sakta hai. Example: SRTF.

CPU-bound Zyada CPU work, comparatively kam I/O.

I/O-bound Zyada disk/network/input wait, comparatively kam CPU work.

Fair scheduler processes ko service deta hai, CPU ko busy rakhta hai, aur response/waiting time ko control karta hai.

CPU scheduling: formulas

Arrival time (AT) = ready queue mein aane ka time. Burst time (BT) = required CPU time. Completion time (CT) = finish time.

Turnaround (TAT) = CT - AT | Waiting (WT) = TAT - BT | Response (RT) = first CPU start - AT

Teen algorithms ko Gantt chart se samjho

Maan lo sab processes t=0 par aaye: P1 ko 5 units, P2 ko 2 units aur P3 ko 1 unit CPU chahiye. Chart mein har block batata hai ki us time CPU kiske paas hai.

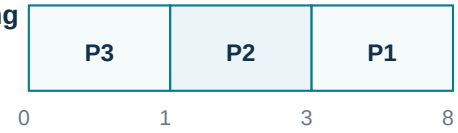
FCFS: arrival order



SJF: shortest first



SRTF: shortest remaining



Note: yahan sab AT = 0, isliye SJF aur SRTF ka order same hai.

Har algorithm ka trade-off

FCFS simple aur starvation-free hai, lekin lamba process chhote processes ko delay kar sakta hai (convoy effect). SJF chhota burst pehle chalata hai; long jobs starve ho sakte hain. SRTF har waqt shortest remaining time leta hai; response achchha ho sakta hai, par context switches zyada hote hain.

Yaad rakho: scheduling choice workload par depend karti hai. Deck ke apne numerical set mein average WT: FCFS 4.75, SJF 4, SRTF 3. Yeh values sirf us example ke liye hain.